

Remove Deprecated Array Comparisons from C++26

Document #: P2865R0
Date: 2023-05-15
Project: Programming Language C++
Audience: Evolution
Revises: N/A
Reply-to: Alisdair Meredith
<ameredith1@bloomberg.net>

Contents

1 Abstract	1
2 Revision history	1
2.1 R0: Varna 2023	1
3 Introduction	2
4 Analysis	2
5 Past Feedback	2
5.1 Initial Review: telecon 2020/06/09	2
6 Status	3
7 Proposed Wording	3
7.1 Wording Intent	3
7.2 Proposed Changes	4
8 Acknowledgements	5
9 References	5

1 Abstract

Comparison operators applied to arrays perform that comparison on the results of array-to-pointer decay, i.e., on the address not the contents of each array. That behavior was deprecated by C++20, and this paper proposes making such comparison ill-formed in C++26.

2 Revision history

2.1 R0: Varna 2023

Initial draft of this paper, based on [\[N4944\]](#).

Notable changes since [\[P2139R2\]](#)

- Added the concern for relational comparison, rather than just equality tests
- Overhauled Core wording, based on [\[N4928\]](#)
- Retested compilers for deprecation warnings

3 Introduction

At the start of the C++23 cycle, [P2139R2] tried to review each deprecated feature of C++ to see which we would benefit from actively removing and which might now be better undeprecated. Consolidating all this analysis into one place was intended to ease the (L)EWG review process but in return gave the author so much feedback that the next revision of the paper was not completed.

For the C++26 cycle, a much shorter paper, [P2863R0], will track the overall analysis, but for features that the author wants to actively progress, a distinct paper will decouple progress from the larger paper so that the delays on a single feature do not hold up progress on all.

This paper takes up the deprecated comparison operators for array types, D.4 [depr.array.comp].

4 Analysis

Comparison of array types was deprecated by C++20 as part of the effort to make the new spaceship operator do the right thing, adopted by paper [P1120R0]. It potentially impacts on code written against C++98 and later standards.

The deprecated comparison operator for arrays was not so much a deliberately designed feature, but accidental oversight that array-to-pointer decay would kick in. Hence, on equality comparison we test whether two arrays are literally the same array at the same address, rather than whether two distinct arrays have the same contents. Identity tests are more typically performed by explicitly taking the address of the objects we wish to validate; it would be highly unusual to rely on an implicit pointer decay to intentionally perform that task. Relying on implicit decay offers no efficiency gain over explicitly taking the address of the array, but would fool a large number of subsequent code readers and reviewers who are not familiar with this idiom. We do note that function comparison performs exactly the same decay, and users are not surprised that comparing functions is an identity test.

The situation is worse for greater or less than comparisons, where the result is often unspecified. The only cases where the result *is* specified are:

- both arrays are the same object (identity test)
- both arrays are data members of the same object
- both arrays are elements of the same array-of-arrays

In all other cases the result is unspecified, as per pointer comparisons, although may be defined for specific ABIs.

Given the likelihood that any usage of these comparison operators is a bug waiting to be detected, this could be a real concern for software reliability. Thus, we recommend removing this feature from C++26.

However, note for wording, that the comparison of an array with a pointer value was not deprecated in C++20, is reasonably idiomatic for practitioners of C++, and is intended to continue to be supported. This mimics the behavior of the spaceship operator.

5 Past Feedback

5.1 Initial Review: telecon 2020/06/09

It was noted that comparison of pointer values is often unspecified, rather than well-defined, unless both arrays happen to be members of the same class, or are both elements of the same multi-dimensional array.

General agreement that this is a good opportunity to remove a landmine from the language that offers little benefit, even when correctly used as intended.

A concern was raised about ongoing C compatibility, which should be forwarded to our WG14 liaison (now sg22), to hopefully have a co-ordinated process for removing this feature.

Polls:

Q1: In C++23, remove the deprecated use of array comparisons?

	SF		F		N		A		SA	
	9		9		3		0		0	

Consensus: Follow up with this direction.

6 Status

To test the current status of this deprecated feature, the following test program was used across a number of compilers and their versions, invoked with the necessary switch to access C++20 features:

```
int main() {
    int a[5] = {};
    int b[5] = {};

    if (a == b) {
        return 1;
    }

    if (a > b) {
        return 2;
    }
}
```

The Clang compiler has been diagnosing warnings in these cases since Clang 3.3 as a regular QoI warning, and with the deprecated notice since Clang 10.

The Microsoft compiler has been warning about the deprecation since MSVC 19.22.

Gcc still does not warn about this deprecation, even in the current trunk build as this paper is written.

Using the classic Intel compiler as a proxy for the EDG front ends, there were no deprecation warnings as of the last release of that compiler, icc 2021.7.1.

The Clang and gcc compilers produced a program that returned 2, whereas the Microsoft and Intel compilers produced a program that returned 0, demonstrating real world differences for the unspecified return value of the comparison.

Note that this status is unchanged since the earlier work on this paper in June 2021.

7 Proposed Wording

All changes are relative to [\[N4944\]](#).

7.1 Wording Intent

The key change is to perform array-to-pointer conversion only if the other operand is a pointer. This makes the distinction between operand and converted operand more important to call out, but mostly for the case of pointer types as none of the conversions will change the type category of an operand unless it is to a pointer type (array-to-pointer conversion and function-to-pointer conversion).

In both subclauses below, the second paragraph has a list of types that the *converted* operand *shall* be, and neither subclause supports array types at that point. Thus, comparison of arrays with anything but pointers is an error, just as any comparison with `void` is an error.

7.2 Proposed Changes

7.6.9 [expr.rel] Relational operators

- ¹ The relational operators group left-to-right.

[Example: `a<b<c` means `(a<b)<c` and *not* `(a<b)&&(b<c)`. —end example]

relational-expression :

compare-expression

relational-expression < *compare-expression*

relational-expression > *compare-expression*

relational-expression <= *compare-expression*

relational-expression >= *compare-expression*

The lvalue-to-rvalue (7.3.2 [conv.lval]), ~~array-to-pointer (7.3.3 [conv.array]),~~ and function-to-pointer (7.3.4 [conv.func]) standard conversions are performed on the operands. If at least one of the operands is a pointer, array-to-pointer conversions (7.3.3 [conv.array]) are performed. ~~The comparison is deprecated if both operands were of array type prior to these conversions (D.4 [depr.array.comp]).~~

- ² The converted operands shall have arithmetic, enumeration, or pointer type. The operators < (less than), > (greater than), <= (less than or equal to), and >= (greater than or equal to) all yield **false** or **true**. The type of the result is **bool**.
- ³ The usual arithmetic conversions (7.4 [expr.arith.conv]) are performed on operands of arithmetic or enumeration type. If both converted operands are pointers, pointer conversions (7.3.12 [conv.ptr]) and qualification conversions (7.3.6 [conv.qual]) are performed to bring them to their composite pointer type (7.2.2 [expr.type]). After conversions, the operands shall have the same type.
- ⁴ The result of comparing unequal pointers to objects ...

7.6.10 [expr.eq] Equality operators

equality-expression :

relational-expression

equality-expression == *relational-expression*

equality-expression != *relational-expression*

- ¹ The == (equal to) and the != (not equal to) operators group left-to-right. The lvalue-to-rvalue (7.3.2 [conv.lval]), ~~array-to-pointer (7.3.3 [conv.array]),~~ and function-to-pointer (7.3.4 [conv.func]) standard conversions are performed on the operands. If at least one of the operands is a pointer, array-to-pointer conversions (7.3.3 [conv.array]) are performed. ~~The comparison is deprecated if both operands were of array type prior to these conversions (D.4 [depr.array.comp]).~~
- ² The converted operands shall have arithmetic, enumeration, pointer, or pointer-to-member type, or type **std::nullptr_t**. The operators == and != both yield **true** or **false**, i.e., a result of type **bool**. In each case below, the operands shall have the same type after the specified conversions have been applied.
- ³ If at least one of the converted operands is a pointer, pointer conversions (7.3.12 [conv.ptr]), function pointer conversions (7.3.14 [conv.fctptr]), and qualification conversions (7.3.6 [conv.qual]) are performed on both operands to bring them to their composite pointer type (7.2.2 [expr.type]). Comparing pointers is defined as follows:
- If one pointer represents the address of a complete object, and another pointer represents the address one past the last element of a different complete object, the result of the comparison is unspecified.
 - Otherwise, if the pointers are both null, both point to the same function, or both represent the same address (6.8.4 [basic.compound]), they compare equal.
 - Otherwise, the pointers compare unequal.
- ⁴ If at least one of the operands is a pointer to member, ...

C.1.X Clause 7: Expressions [diff.cpp23.expr]

Affected subclause: 7.6.9 [expr.rel] and 7.6.10 [expr.eq]

Change: Cannot compare two objects of array type.

Rationale: The old behavior was confusing, as it did not compare the contents of the two arrays, but compare their addresses. Depending on context, this would either report whether the two arrays were the same object, or have an unspecified result.

Effect on original feature: A valid C++ 2023 program directly comparing two array objects will be rejected as ill-formed in this International Standard. [Example 1:

```
int arr1[5];
int arr2[5];
bool same = arr1 == arr2;           // ill-formed; previously well-formed
bool idem = arr1 == +arr2;          // compare addresses, unspecified result
```

—end example]

C.5.3 Clause 7: expressions [diff.expr]

Affected subclause: 7.6.9 [expr.rel] and 7.6.10 [expr.eq]

Change: Cannot compare two objects of array type.

Rationale: The behavior was confusing, as it did not compare the contents of the two arrays, but compared their addresses. Depending on context, this would either report whether the two arrays were the same object, or have an unspecified result.

Effect on original feature: Deletion of feature with unspecified behavior.

Difficulty of converting: Violations will be diagnosed by the C++ translator.

How widely used: Rare. In the cases where the result is well defined, it is reporting whether both arguments are the same object, using the same name.

D.4 [depr.array.comp] Array comparisons

- ¹ Equality and relational comparisons (7.6.10 [expr.eq], 7.6.9 [expr.rel]) between two operands of array type are deprecated.

[Note 1: Three-way comparisons (7.6.8 [expr.spaceship]) between such operands are ill-formed. —end note]

[Example 1:

```
int arr1[5];
int arr2[5];
bool same = arr1 == arr2;           // deprecated, same as &arr1[0] == &arr2[0],
                                   // does not compare array contents
auto cmp = arr1 <=> arr2;           // error
```

—end example]

8 Acknowledgements

Thanks to Michael Parks for the pandoc-based framework used to transform this document's source from Markdown.

Thanks again to Matt Godbolt for maintaining Compiler Explorer, the best public resource for C++ compiler and library archaeology, especially when researching the history of deprecation warnings!

9 References

- [N4928] Thomas Köppe. 2022-12-18. Working Draft, Standard for Programming Language C++. <https://wg21.link/n4928>

- [N4944] Thomas Köppe. 2023-03-22. Working Draft, Standard for Programming Language C++. <https://wg21.link/n4944>
- [P1120R0] Richard Smith. 2018-06-08. Consistency improvements for `<=>` and other comparison operators. <https://wg21.link/p1120r0>
- [P2139R2] Alisdair Meredith. 2020-07-15. Reviewing Deprecated Facilities of C++20 for C++23. <https://wg21.link/p2139r2>